

Monetary transmission in a fragmented, uncertain world*

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Abstract

This paper investigates how economic uncertainty affects monetary policy transmission using Bayesian Threshold VARs across 30 economies. We find that during high uncertainty, the inflation response to monetary policy shocks is markedly weaker, while the GDP response remains stable. Consequently, state-dependent sacrifice ratios show disinflation is significantly more costly when uncertainty is elevated. The asymmetry we find is primarily driven by policy uncertainty and suggests that this uncertainty disrupts transmission by weakening the pass-through from economic slack to inflation.

Keywords: Monetary transmission, Economic uncertainty, Threshold Vector Autoregression, Sign restrictions

JEL: E32, E44, E52, C32

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Introduction

How monetary policy is transmitted to the real economy is one of the oldest and most consequential questions in macroeconomics. Understanding these transmission dynamics has taken on renewed importance in an environment marked by unusually frequent and persistent uncertainty. Importantly, this environment is best characterized not by calculable risk, but by what Dequech (2000) terms fundamental uncertainty: a state in which the future is inherently uncreated and unknowable, rather than merely obscured by missing information. Today, this fundamental uncertainty is especially pronounced against the backdrop of geoeconomic fragmentation (the reconfiguration of trade, investment, and technology relationships along geopolitical rather than purely economic lines), which, as Ilyina et al. (2023) point out, continually generates structural economic and policy shocks.

This reconfiguration has accelerated markedly since the COVID-19 pandemic first exposed the fragility of global supply chains. Russia’s invasion of Ukraine in 2022 triggered a large-scale sanctions regime and a near-collapse of European gas imports from Russia, feeding directly into the surge in euro area inflation that followed and prompting one of the fastest monetary tightening cycles in the ECB’s history. Export controls on semiconductors and critical minerals, alongside a broader push toward “friend-shoring” and reshoring of production, have reshaped US-China economic relations since 2022, while global tariffs rose sharply again in 2024–2025 amid a fresh round of trade-policy escalation. Measures of policy and trade uncertainty have surged in step with these developments, and by 2025 several indicators of geopolitical and trade fragmentation stood at or near multi-decade highs (see, e.g., the ECB/ESRB (2026) report).

These developments naturally interact with monetary policy transmission. As Andriantomanga et al. (2022) and Bolhuis et al. (2023) note, geoeconomic fragmentation carries substantial real economic costs and heightened inflationary pressures, both of which central banks must navigate. Crucially, when operating under deep fundamental uncertainty, economic agents may shift away from rational optimization toward the realm of Keynesian “animal spirits” — the gut feelings, emotions, and spontaneous optimism that drive human economic decisions when future outcomes cannot be mathematically calculated (Akerlof and Shiller (2010)). Understanding how monetary policy operates in this behavioral realm is vital, as conventional interest rate signals may become distorted.

In order to accurately assess the efficacy of central bank policy in achieving its mandates under high uncertainty, models must account for these structural and behavioral shifts in the economy’s underlying dynamics.

Given the extensive literature on uncertainty shocks and the broad consensus about their real effects, the immediate implications of heightened uncertainty are comparatively well understood and offer useful guidance for methodological choices. A small but growing literature has begun to address this question directly. This literature tends to find that monetary policy may be less effective in times of high uncertainty. These findings are attributed to the fact that high uncertainty already dampens investment demand and increases consumer precaution; as a result, the demand-pull effect of monetary policy is smaller.

Although these findings provide a consistent narrative across the existing literature, several methodological limitations offer room for extension. Some of the existing frameworks impose restrictive assumptions on how uncertainty alters macroeconomic dynamics, typically allowing only partial structural shifts. For instance, Aastveit et al. (2017) restrict regime dependence strictly to interest rate elasticities, while Aquino et al. (2022) assume that only the monetary policy reaction function shifts during uncertain times. Such constraints may fail to capture the full spectrum of state-dependent interactions across the economy.

Other empirical findings often rely on specific proxies or older samples that may not fully reflect contemporary, globalized fundamental uncertainty. While Castelnovo and Pellegrino (2018) document a significantly milder real effect of unexpected policy moves when macroeconomic uncertainty is high, their sample concludes in 2008, predating the Global Financial Crisis (GFC). Furthermore, their analysis utilizes a forecast-error-based uncertainty measure (Jurado et al. (2015)), which is conceptually distinct from the modern text- and narrative-based measures widely employed today to capture perceived fundamental uncertainty. Finally, the overwhelming majority of these studies are heavily US-centric; as such, they do not capture whether varying structural characteristics of the economy produce any changes in how uncertainty and monetary policy interact.

Motivated by these considerations, we develop a parsimonious econometric framework to study the relationship between uncertainty and monetary transmission across a broad

panel of countries. Rather than relying on a theory-consistent structural model that is calibrated to match the results of earlier work, we follow the data-driven VAR tradition initiated by Sims (1980) and let the reduced-form dynamics of the data speak. Specifically, we employ a two-regime Threshold Vector Autoregression (TVAR) capable of flexibly capturing state-dependent transmission mechanisms and varying shock impacts across high- and low-uncertainty regimes.

To limit the number of parameters and keep the computational workload in check, the model is deliberately parsimonious, containing only a compact information set intended to capture open-economy dynamics amid uncertainty shocks. Accordingly, our endogenous variables include the policy interest rate, GDP growth, CPI inflation, the real effective exchange rate, and a measure of economic uncertainty. We identify monetary policy shocks using sign restrictions, as this approach does not rely on external high-frequency instruments, which are unavailable for many of the economies included in our broad sample of advanced and emerging markets.

Our results diverge from much of this literature in an important respect. Rather than finding that monetary policy loses its grip on real activity under high uncertainty, as Aastveit et al. (2017), Aquino et al. (2022), and Castelnuovo and Pellegrino (2018) variously do, we find that the response of GDP growth to a policy shock is broadly unchanged across uncertainty regimes. What changes rather, is how much of that contraction shows up as disinflation. Inflation responds less persistently to a monetary tightening when uncertainty is high, so that the sacrifice ratio (the amount of output that must be forgone per unit of disinflation) rises sharply in high-uncertainty periods relative to low-uncertainty ones. This pattern is stable across our broad country panel, across advanced and emerging economies alike, and across alternative definitions of the uncertainty threshold, though it is considerably more pronounced for narrative measures of economic policy uncertainty than for a model-based uncertainty index or a measure of geopolitical risk specifically. We interpret this as evidence that uncertainty disrupts monetary transmission primarily on the inflation side of the mechanism, rather than by weakening the real effects of a policy shock as much of the existing single-country evidence suggests.

The remainder of the paper is organized as follows. The next section reviews the relevant literature. Section three describes the data used in the estimations, with particular emphasis on measures of uncertainty. Section four outlines the methods, including the

estimation of the TVAR with Bayesian methods, the identification strategy as well as further calculations and aggregating assumptions. Section five discusses the results in detail, and section six concludes.

Literature review

We identify three strands of literature that guide our investigation: (i) the real economic effects of uncertainty shocks, (ii) the state-dependence of macroeconomic dynamics (particularly concerning uncertainty), and (iii) how monetary transmission itself varies conditional on the state of the economy. We highlight key contributions from each strand below.

Real effects of uncertainty shocks. A substantial literature finds that uncertainty shocks act much like negative demand shocks, primarily by depressing investment as firms adopt a more precautionary stance (Bloom et al. (2007); Bloom (2009); Bloom (2014); Gilchrist et al. (2014)). Caldara et al. (2016) emphasize the joint role of financial and uncertainty shocks in driving business-cycle fluctuations and argue that the Great Recession likely reflected the interaction of both. Bonciani and Ricci (2020) use local projections to estimate the impact of global financial uncertainty on small open economies and find contractionary effects that are stronger in more open economies and during downturns. Colombo (2013) shows that US uncertainty shocks spill over substantially to the euro area business cycle, with US shocks generally mattering more than euro-area shocks for that spillover.

The real effects of uncertainty also appear to differ systematically between advanced and emerging economies. Carrière-Swallow and Céspedes (2013) show that uncertainty shocks exert greater influence in emerging economies owing to tighter credit constraints, a finding echoed by Ahir et al. (2019). Cheng and Chiu (2018) show that geopolitical risk shocks are important drivers of business cycles in emerging markets specifically, while Jung et al. (2021) use Korean data to show that geopolitical risk triggers sharp declines in stock prices, particularly for large, domestically owned, or asset-heavy firms.

Caggiano and Castelnovo (2021) estimate a global financial uncertainty factor from volatility data spanning 42 countries and find that a meaningful share of the world out-

put loss during the Great Recession would have been avoided in the absence of global uncertainty shocks. Bobasu et al. (2024) build on the measurement approach of Jurado et al. (2015) to construct a global uncertainty measure for the euro area’s main trading partners and show, using a proxy-SVAR instrumented with gold-price movements, that global uncertainty shocks have historically been an important driver of euro area industrial production.

State-dependence in economic dynamics and uncertainty. Another strand of literature investigates whether the dynamics governing the economy’s evolution vary with the state of the economy. Using a smooth-transition VAR, Paccagnini and Colombo (2020) find uncertainty shocks to be significantly more severe in recessions than in normal times, and that central bank balance sheet tools are particularly helpful in mitigating their impact. Alessandri and Mumtaz (2019), using a Threshold VAR, find a similar pattern tied to financial conditions. Uncertainty shocks are inflationary in normal times but deflationary, and more contractionary for output, during periods of financial stress. De Santis and Cardamone (2026) show that the parameters of the Phillips curve, the dynamic IS equation, and the Taylor rule are themselves regime-dependent across joint states of inflation and the output gap, with the size of monetary policy shocks significantly larger when inflation exceeds target.

State-dependence in the transmission of monetary policy. Monetary policy studies using nonlinear models offer some relevant findings as well. Kakes (1998) uses a Markov-switching model and finds monetary policy to be more effective in recessions in both the US and Germany, while Avdjiev and Zeng (2014) find that the impact of monetary shocks is amplified during periods of low growth, even as policy itself tends to be more aggressive during booms. Pellegrino et al. (2023) document, with a nonlinear proxy-VAR, that a large financial uncertainty shock can account for a substantial share of the US output loss during the Great Recession, while systematic monetary policy played a powerful role in limiting that loss. Caggiano et al. (2017) find that the Federal Reserve’s systematic response to uncertainty shocks is considerably more effective at stabilizing real activity in expansions than in recessions.

Discussed more thoroughly in the introduction, Aastveit et al. (2017) and Aquino et al. (2022), using interacted VARs, provide evidence that the effect of monetary policy is dampened by high uncertainty. Using a full TVAR, Castelnovo and Pellegrino (2018) also

find that high macro-uncertainty dampens the transmission, a finding they also rationalize via a New Keynesian model.

Taken together, these findings reinforce that economic uncertainty is a primary driver of global business cycles, with considerable spillovers from global or systemically important economies to smaller ones. The literature also clearly establishes that high financial stress and uncertainty fundamentally alter macroeconomic dynamics. This structural shift affects both the systematic reaction function of monetary policy and the real economy’s responsiveness to policy shocks. For instance, central banks may react directly to uncertainty proxies, or their response to inflation and demand shocks may change conditional on the prevailing level of uncertainty. Concurrently, the state of the economy can dictate the risk appetite and price-setting behavior of agents, subsequently steepening or flattening the slopes of the Phillips and IS curves.

Finally, the empirical definition of “good” versus “bad” times can drastically alter conclusions: as highlighted above, “bad times” defined by one metric (e.g., low growth) may amplify monetary policy effectiveness, whereas “bad times” defined by another (e.g., high uncertainty) may dampen it. We aim to contribute to this literature by systematically testing how high uncertainty measured across several distinct indicators interacts with the transmission of interest rate shocks across a large cross-section of economies.

Data

Macro aggregates. Our empirical analysis relies on a compact set of four core macroeconomic variables¹: real GDP, consumer price inflation, the real effective exchange rate and the policy interest rate. The data is sampled at a quarterly frequency covering the period from 1995Q1 to 2026Q1. The resulting panel is unbalanced, as data for certain emerging economies only becomes available in the early 2000s. We view this compact set of variables as the necessary minimum to adequately capture open-economy macroeconomic dynamics. We choose to keep the model parsimonious in order to be able to include

¹All macro aggregates except the policy interest rate are expressed in year-over-year growth rates. Data for real GDP is retrieved from the IMF’s database, while the rest of the series are retrieved from BIS data. Inflation and GDP data for the Euro Area are retrieved from Eurostat, while data for the effective exchange rate is retrieved from the ECB’s data portal, and the Euro Area policy interest rate and shadow rate come from the database of Krippner (2013).

a relatively large panel of economies in our analysis. Further, maintaining a parsimonious model mitigates the substantial computational workload of estimating regime-switching models with Bayesian methods over a large cross-section of countries.

The estimation is carried out individually across a diverse panel of 30 economies, encompassing a wide range of large, small, advanced, and emerging markets.² Notably, rather than modeling individual member states of the European Monetary Union, we incorporate them via a single Euro Area³ aggregate. Since all member states are subject to a common monetary policy stance, this aggregation choice is consistent with our core objective of analyzing the transmission of monetary policy shocks. Additionally, replacing individual members of the Euro Area with a single aggregate substantially reduces overall estimation time. Whether ECB monetary policy impacts individual member countries differently in high- versus low-uncertainty times is left for future research.

In order to account for structural change across the decades, we apply two essential transformations to the data to ensure robust structural identification. First, to account for low-frequency shifts in inflation targets and evolving monetary policy regimes over our three-decade sample, we detrend both the inflation and interest rate series using the regression-based filter proposed by Hamilton (2018). Second, to accurately capture the unconstrained stance of monetary policy during episodes when nominal rates hit the zero lower bound (ZLB), we substitute the observed policy rates with shadow short rates wherever applicable. Specifically, we utilize the shadow rate framework developed by Krippner (2013)⁴, supplemented by the shadow rate estimates of De Rezende and Ristiniemi (2023) for Sweden.

Uncertainty measures. Because economic uncertainty is inherently unobservable, quantifying it requires proxy indicators. The macroeconomic literature has traditionally relied on financial market metrics, such as implied volatility indices (most commonly the VIX), equity market volatility (such as the EMV tracker from Baker et al. (2026)), or broader financial stress indicators (such as the Chicago Fed’s NFCI or the ECB’s CLIFS).

²The sample includes Australia, Brazil, Canada, Switzerland, Chile, Colombia, the Czech Republic, Denmark, the Euro Area, the United Kingdom, Hungary, Indonesia, India, Iceland, Israel, Japan, South Korea, Mexico, North Macedonia, Norway, New Zealand, the Philippines, Poland, Romania, Singapore, Serbia, Sweden, Thailand, the United States, and South Africa.

³We use the “Euro Area (changing composition)” definition of Eurostat and the ECB to account for new entrants over the decades.

⁴The dataset covers Australia, Canada, Switzerland, the Euro Area, the United Kingdom, Japan, New Zealand, Singapore and the United States.

While highly responsive, these market-based metrics primarily capture the risk premia and volatility expectations of financial market participants. Consequently, they often conflate true macroeconomic uncertainty with financial stress and asset price imbalances which may not accurately reflect broader uncertainty that households and firms face. Following the work of Baker et al. (2016), text query-based indices have emerged as a robust alternative to gauging macro-level uncertainty instead of proxying it via financial market-based measures. By scraping news archives and policy reports for specific keywords, these narrative-based measures capture a more direct, qualitative reflection of uncertainty as it is actually perceived and articulated by economic agents, policymakers, and journalists in real time.

Under such considerations, we also choose to use such an index for the bulk of our analysis. Namely, we use the World Uncertainty Index (WUI) developed by Ahir et al. (2022). We select the WUI as our baseline for several reasons. First, it is based on IMF Economist Intelligence Unit quarterly reports which are compiled by IMF staff. As such, the underlying text from which the index is built is created by a highly reliable source and it has global coverage. Second, the search terms used to compile the index offer a very broad definition of economic uncertainty, as the index merely counts the number of times the word “uncertainty” and its variants occur in each report.

In order to ensure our findings are not driven by the idiosyncrasies of a single indicator, we employ several alternative metrics as sensitivity checks. These include the US Economic Policy Uncertainty (EPU) index of Baker et al. (2016), the 1-month ahead macroeconomic uncertainty index (JLN1M) of Jurado et al. (2015), and the global Geopolitical Risk (GPR) index of Caldara and Iacoviello (2022). Although the EPU and the JLN1M are derived from US data rather than global data, we choose the original US-specific versions, as many economic policy and macrofinancial uncertainty episodes (and the global spillovers that follow) frequently originate from the United States. As such, while these may not be truly global measures of uncertainty, they serve as close proxies of global uncertainty processes.

It is also important to note the conceptual nuances among these proxies. The index proposed by Jurado et al. (2015) is derived econometrically from the unforecastable components of a broad set of macroeconomic indicators. As such, it conceptually lies somewhat closer to volatility and financial stress measures than to the text-based uncertainty mea-

sure counterparts. On the other hand, while the GPR index of Caldara and Iacoviello (2022) follows the text-based measurement of the other indices, its search terms focus more on geopolitical tension rather than strictly economic (policy) uncertainty. We still choose to include this index in our analysis because empirical evidence overwhelmingly demonstrates that geopolitical shocks trigger macroeconomic contractions that closely mirror pure uncertainty shocks.

Finally, we apply necessary transformations to these indices prior to estimation. Due to the apparent upward sloping trends in the narrative series, we detrend both the WUI and the EPU indices also using the regression-based filter of Hamilton (2018). In contrast, the GPR and JLN1M indices exhibit stable long-run means and are therefore simply demeaned. In our baseline specification, we define a “high uncertainty” regime as any period where the chosen index lies above its historical median. We also conduct two sensitivity exercises regarding this threshold. We raise the cutoff to the 70th percentile of the historical distribution to investigate whether more extreme levels of uncertainty distort transmission dynamics further, and we also classify regimes based on “cyclical” uncertainty, where a high regime is strictly defined as periods when the indicator sits above its trend (mean). The uncertainty indices and their respective baseline regime classifications are plotted in detrended (demeaned) levels in Figure 1. In all VAR specifications, the chosen uncertainty measure enters as the fifth endogenous variable and is scaled to a unit standard deviation.

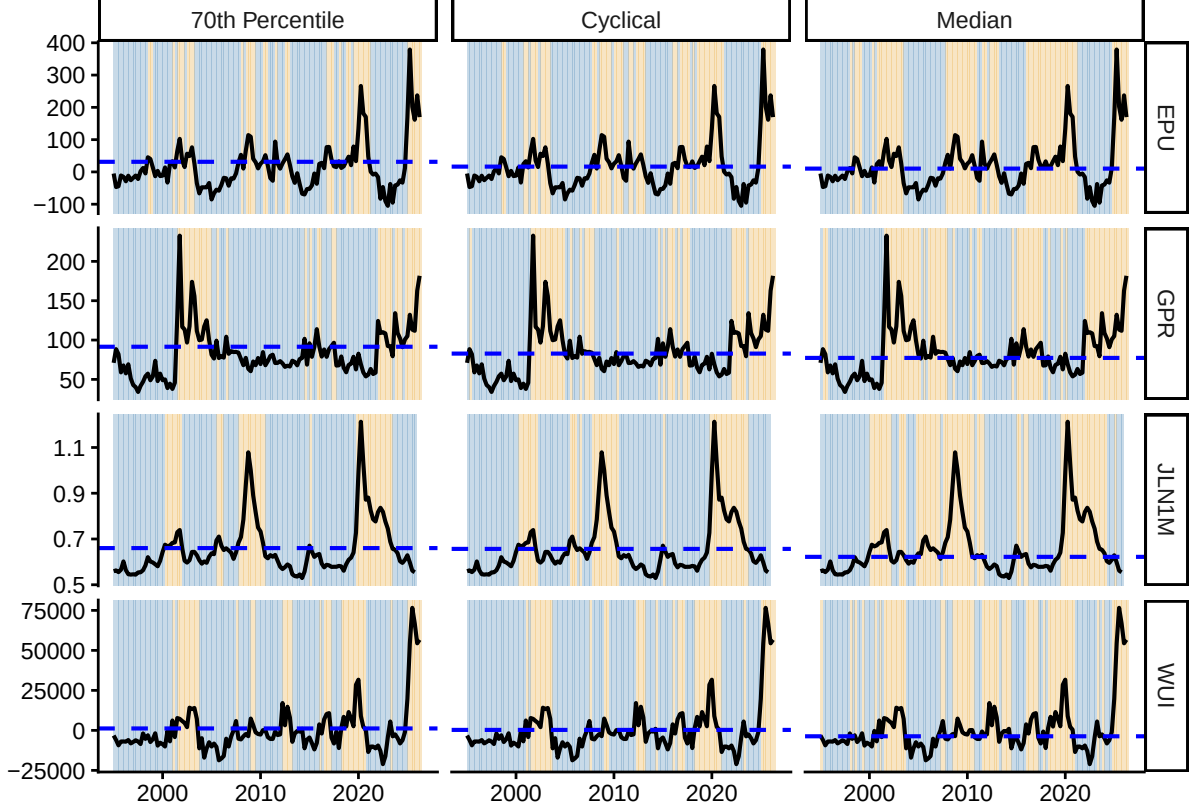


Figure 1: Uncertainty measures and regime classifications.

Notes: The dashed blue lines indicate the threshold levels. Blue shading and yellow shading along the series indicates the low uncertainty and high uncertainty regimes respectively.

Empirical framework

Bayesian Threshold VAR estimation. Our analysis is based on a compact structural threshold vector autoregression (TVAR) with five endogenous variables: GDP growth, inflation, the effective exchange rate, the nominal interest rate, and a measure of global uncertainty. In all cases, the TVAR is estimated with two regimes, corresponding to periods of high and low uncertainty. Mathematically, the model is specified as:

$$Y_t = \sum_{k=1}^4 \Theta_{high,k} I(X_{t-1} \geq \mu) Y_{t-k} + \sum_{k=1}^4 \Theta_{low,k} I(X_{t-1} < \mu) Y_{t-k} + u_t \quad (1)$$

where Y_t is the vector of endogenous variables, $\Theta_{low,k}$ and $\Theta_{high,k}$ are the coefficient matrices for each regime, $I(.)$ is the regime indicator function, X_{t-1} is the (one-quarter lag of the) threshold variable, μ is the threshold value and u_t is the error term.

We estimate these models for each country in the sample individually and subsequently aggregate the resulting impulse responses post estimation. All models are estimated using Bayesian methods following Giannone et al. (2015). While the VAR coefficients (and contemporaneous structural responses) are allowed to be regime-specific, we simplify the prior specification by setting the overall degree of shrinkage equal to that of a linear VAR estimated on the full sample. This strategy ensures that any differences between impulse responses of the two states are caused by differences in dynamics and not by asymmetric or uneven priors. This simplification is done in order to streamline the interpretation of the results, though it leaves the question of whether prior shrinkage should be regime-specific to future research. All models are estimated using 4 lags and 5,000 posterior draws.

Impulse responses and aggregation. To synthesize our findings and evaluate representative structural dynamics across the panel, we implement a posterior pooling strategy. For our baseline results, we assume that the country-specific distributions of IRFs are independent draws from the global (representative) IRFs. Under this assumption, we calculate the median and 68% credibility sets from a distribution that stacks draws across all countries in our sample. By aggregating the posterior distributions in this manner, we capture the central tendency of the global transmission mechanisms while fully preserving the underlying estimation uncertainty from the individual country models.

Importantly, we report purely regime-specific rather than generalized impulse response functions. These impulse responses reflect the dynamics as if the economy remained in the regime where the shock originated. We view this assumption as economically acceptable for two primary reasons. First, the core objective of our investigation is precisely to isolate and evaluate how a shock propagates within a specific state of uncertainty. Second, these uncertainty regimes are relatively stable over time, as demonstrated in Figure 1, which minimizes the likelihood of rapid regime-switching during the transmission horizon. In this spirit, our methodological approach closely aligns with the regime-specific transmission analysis recently employed by De Santis and Cardamone (2026).

Beyond the baseline global aggregation, we also check for heterogeneity across different

country groups to ascertain whether there are systematic differences in how uncertainty influences the transmission of monetary policy along various structural characteristics. When evaluating these subsets (such as advanced versus emerging market economies), we apply similar aggregating assumptions. For example, when separating the panel by development level, we assume that the country-specific distributions are independent draws from the global advanced economy IRFs and the global emerging market economy IRFs, respectively. As such, we stack the posterior draws within each subgroup and extract the representative set of IRFs. Repeating this approach along several structural characteristics allows us to check which ones improve or deteriorate the efficacy of monetary policy in high-uncertainty times. We provide an extensive list of these characteristics and related country groupings in the Appendix.

Identification. We identify shocks using sign restrictions, following the methodology of Arias et al. (2018). To properly isolate the monetary policy shock, we identify four additional structural shocks in the model: a domestic demand shock, a cost-push supply shock, a global uncertainty shock, and a risk premium (exchange rate appreciation) shock. Additionally, we leverage the well-documented transmission lags of monetary policy (see, e.g., Friedman (1961) and Batini and Nelson (2001)) and place restrictions for this shock to horizons $h = 1, 2$ quarters ahead as well. This is implemented as an extra precaution in isolating the monetary policy shock from any other possible structural shocks that the small variable set may not cover. The sign restrictions used for the identification are summarized in Tables 1 and 2 below.

Forni and Gambetti (2014) show that a VAR model can fail to recover the true structural shocks if it does not adequately span the economy’s relevant information set. Due to the large cross-section of countries in our panel, we do not formally test for informational sufficiency, although such a test could, in principle, be conducted on a country-by-country basis. We believe that our five endogenous variables provide adequate coverage to capture the primary shocks faced by an average open economy. Namely, domestic demand, cost-push, monetary policy, external risk premium, and global uncertainty⁵ shocks. We

⁵Caldara and Herbst (2019) for instance show that correctly identifying monetary shocks requires explicitly modelling the systematic reaction of policy to credit spreads. Failing to do so induces an attenuation bias that can itself masquerade as weak transmission. While we do not possess credit spread data, adding the endogenous (and regime-specific) reaction to uncertainty should help sharpen the identification of monetary policy shocks.

	UIP	Demand	Cost-push	Monetary Policy	Global uncertainty
EER	+			+	
GDP	-	+	-	-	-
π	-	+	+	-	-
R	-	+	+	+	
U					+

Table 1: Sign restrictions, $h = 0$

	UIP	Demand	Cost-push	Monetary Policy	Global uncertainty
EER					
GDP				-	
π				-	
R				+	
U					

Table 2: Sign restrictions, $h = 1, 2$

acknowledge that a richer dataset incorporating factors such as detailed commodity price dynamics, credit constraints, financial frictions, technological innovations, and fiscal variables could certainly sharpen the identification. Doing so, however, would necessitate dropping several countries from the sample due to data availability and would impose a significantly heavier computational burden. As such, we leave these expanded specifications for future research. Nevertheless, to ensure the reliability of our baseline findings, we do relax the persistence assumption on monetary policy shocks in our robustness exercises.

The economic reasoning behind these identification restrictions is grounded in standard macroeconomic theory. First, we leave global uncertainty unconstrained across all domestic shocks. This reflects the standard small open economy assumption: it remains ambiguous whether individual countries of the sample carry sufficient weight in international dynamics to significantly impact global uncertainty.

Second, we separate domestic demand shocks from global uncertainty shocks. A positive domestic demand shock simultaneously raises output, inflation ($GDP \uparrow, \pi \uparrow$) and induces an endogenous monetary policy tightening ($R \uparrow$). A global uncertainty shock similarly acts as an aggregate demand contraction on domestic real activity ($GDP \downarrow, \pi \downarrow$), but it is uniquely identified by an exogenous spike in global uncertainty ($U \uparrow$) and an ambiguous response of interest rates. Leaving R unconstrained accounts for the ambiguity in how monetary policy might respond to uncertainty. Following the Knightian uncertainty nar-

rative, it might trigger a “wait-and-see” type response. Alternatively, global uncertainty spikes might lead to capital flight-to-safety or currency depreciation, which might force the central bank to balance domestic easing against defensive rate hikes.

Third, the cost-push supply shock captures standard stagflationary dynamics, driving output down while pushing inflation up ($GDP \downarrow, \pi \uparrow$). Monetary authorities are assumed to react systematically by raising policy rates ($R \uparrow$) to control the inflationary pressure. Finally, the UIP shock isolates exogenous currency appreciation ($EER \uparrow$) driven by risk premium shifts (or international capital flows). The resulting loss of price competitiveness and cheaper imported inputs subsequently compress domestic output and consumer prices ($GDP \downarrow, \pi \downarrow$). To cleanly disentangle this from a monetary policy contraction, we impose that the central bank leans against the appreciating exchange rate and eases interest rates ($R \downarrow$).

Measuring monetary policy efficacy. When monetary policy intervenes to curb inflation, it inherently comes with a cost in terms of real output losses because it pulls back aggregate demand. Traditionally, the sacrifice ratio serves as a measure to gauge how much cumulative loss in output the economy must endure to achieve a given reduction in inflation. In the context of our VAR framework, we construct this metric at each horizon h by taking the ratio of the cumulative impulse response of GDP growth to the cumulative impulse response of inflation. Mathematically, the sacrifice ratio IRFs are defined as

$$SR_h = \frac{\sum_{h=0}^H IRF_h^{GDP}}{\sum_{h=0}^H IRF_h^{\pi}} \quad (2)$$

where IRF_h^{GDP} and IRF_h^{π} denote the orthogonalized impulse responses of GDP growth and inflation to a contractionary monetary policy shock at horizon h . As such, this measure provides a standardized measure of how much real growth needs to be sacrificed per unit of disinflation. Besides standardizing the results (i.e. independent of the size of the monetary policy shock), reporting sacrifice ratio IRFs helps us compress information into a few easily interpretable graphs that describe how the efficiency of the transmission changes when uncertainty is high.

To properly account for estimation uncertainty, the sacrifice ratio is not calculated simply by dividing the median impulse responses. Instead, the ratio in Equation 2 is computed

dynamically at each individual posterior draw. From this resulting posterior distribution of the sacrifice ratios, we extract the posterior medians to be presented in the main figures. Aggregation of the sacrifice ratio IRFs follows the same aggregation logic outlined before. It should be noted that while the sacrifice ratio IRFs help streamline the interpretation of the results, using them as the main visual tool has a downside. Because the sacrifice ratio requires dividing two estimated, normally distributed random variables, the resulting ratio-distribution is heavy tailed and can produce highly variable, possibly extreme and asymmetric density functions (see, e.g., Yang and Gui (2023)). Subsequently, wide 68% credibility sets would obscure the central tendencies and visual interpretation of the results. As such, for all sacrifice ratio IRFs in the main text, we report the medians exclusively. We report some of the sacrifice ratio IRFs with the 68% credibility set in the Appendix in order to fully account for the estimation uncertainty of these impulse responses as well.

Results

Figure 2 reports the central result of the paper. While the impulse responses of GDP growth, the policy rate, and the exchange rate to a monetary policy shock are broadly similar across uncertainty regimes, the response of inflation is not, as it is less persistent in the high-uncertainty regime than in the low-uncertainty regime. The implied sacrifice ratio (Panel B) makes this concrete. After the initial transition, roughly one unit of cumulative output loss is needed per unit of disinflation in the high-uncertainty regime, versus a small fraction of a unit in the low-uncertainty regime. High uncertainty does not so much prevent monetary policy from contracting the economy as it prevents that contraction from translating into disinflation.

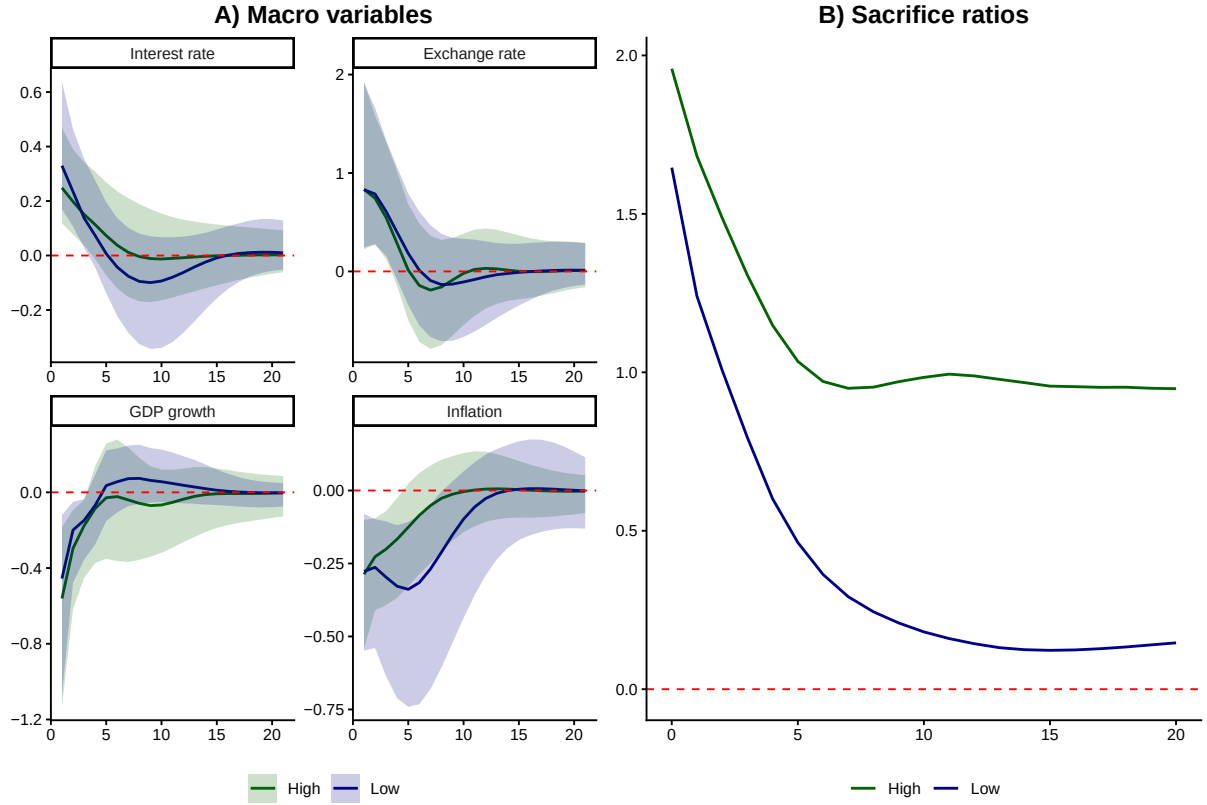


Figure 2: State-dependent impulse responses to a monetary policy shock.

Notes: Green (High) and blue (Low) lines show posterior median impulse responses in the high- and low-uncertainty regimes, respectively, pooled across all sample countries; shaded bands show 68% credibility sets. Panel B reports the implied sacrifice ratio, defined as the ratio of the cumulative impulse response of GDP growth to that of inflation (Equation 2).

This differs from the mechanism emphasized in much of the earlier literature, where uncertainty dampens transmission primarily by muting the real response to a policy shock (e.g., Castelnuovo and Pellegrino (2018)). In our setting, GDP contracts by a broadly comparable amount in both regimes. What changes is how much of that contraction shows up as disinflation. Read together with Figures 8 and 9 in the Appendix, the pattern is consistent across both the full sample and the shadow-rate subsample. The nonlinearity is primarily reflected in how inflation responds to the policy shock.

While the individual impulse responses (taking all shocks into account in Figures 8 and 9 in the Appendix) look broadly similar in both regimes, this does not invalidate the use of the threshold model. A modest, consistent gap in the cumulative inflation response

compounds over the horizon into a large gap in the sacrifice ratio (as seen before in Panel B of Figure 2 before). A linear model that pooled both regimes would report a single blended inflation response that correctly describes neither one, and would misstate the sacrifice ratio in both states⁶. The fact that this separation survives across a barrage of sensitivity tests further suggests a stable underlying feature of the data rather than an artifact of any one specification.

The stark divergence in price dynamics across uncertainty regimes could suggest a flattening of the Phillips Curve under high economic policy uncertainty. In the presence of nominal rigidities, elevated policy uncertainty increases the real option value of waiting, prompting the “wait-and-see” type response widely documented in uncertainty literature as well. As forecasting future marginal costs and aggregate demand becomes increasingly difficult, firms systematically opt against costly nominal price markdowns. Consequently, less of a given change in economic slack passes through into prices. This accounts for both the shallower inflation trough and its faster reversion in the high-uncertainty regime.

Alternatively, the pattern could be driven by how economic agents form inflation expectations during volatile periods. Under high uncertainty, agents face a harder signal-extraction problem; with more disturbances in play, it is rational to attribute less of any given inflation surprise to a persistent shift, and thus they revise their expectations by less. This mechanically flattens the backward-looking empirical Phillips curve without requiring any structural change in firms’ underlying price-setting behavior. Furthermore, if agents perceive severe policy uncertainty as a precursor to adverse supply shocks or eventual fiscal dominance, their inflation expectations may retain an upward bias, stubbornly anchoring pricing behavior and directly counteracting the intended disinflationary pressures of the interest rate hike.

Because our reduced-form threshold VAR identifies shocks via sign restrictions rather than an explicit structural model, we cannot definitively distinguish between a change in price-setting behavior and a shift in the anchoring of expectations. We present both as plausible, non-competing candidates. We leave the direct more precise identification between these channels to future research utilizing structural models. We however do also observe a “wait-and-see” type reaction in the policy rate path as well, as under

⁶Figure 7 reports the linear BVAR impulse responses pooled across our panel. By inspecting the impulse responses to a monetary policy shock, we can visually confirm that the linear impulse response of inflation appears indeed to be a blend of the two regimes’ responses.

low uncertainty the stance appears to normalize more aggressively. This, as well as the consensus of the uncertainty literature on “wait-and-see” motives prompts us to slightly favor this narrative.

How much do country characteristics influence the results? While Figure 2 establishes the average pattern, Figure 3 asks whether structural country characteristics⁷ (factors outside the estimated system itself) help explain which economies bear a larger or smaller uncertainty penalty. The results established in Figure 2 are remarkably stable. Across every partition we consider, the high-uncertainty sacrifice ratio remains well above the low-uncertainty one, so none of these characteristics overturns the headline result. Where characteristics matter, they matter mostly at the margin, and mostly within the high-uncertainty regime.

Advanced economies face a somewhat worse sacrifice ratio than emerging markets under high uncertainty, the opposite of what a standard credit-constraint story for emerging-market vulnerability would predict. This at face value is a surprising result, however we attribute this to other factors being highly correlated with the EME classification. For instance, Panel E reports that economies with high trade openness face a more favorable sacrifice ratio compared to the global one, possibly due to international diversification and risk sharing. Net commodity exporters likewise fare worse than net importers under high uncertainty, consistent with commodity-driven inflation being less responsive to domestic demand conditions specifically when uncertainty is elevated. This is a pattern that falls in line with the aggregate-supply interpretation of the main result above, since commodity-price pass-through operates largely outside the domestic output gap altogether.

⁷We provide details on how countries assigned into each subgroup along each of the structural characteristics in the Appendix.

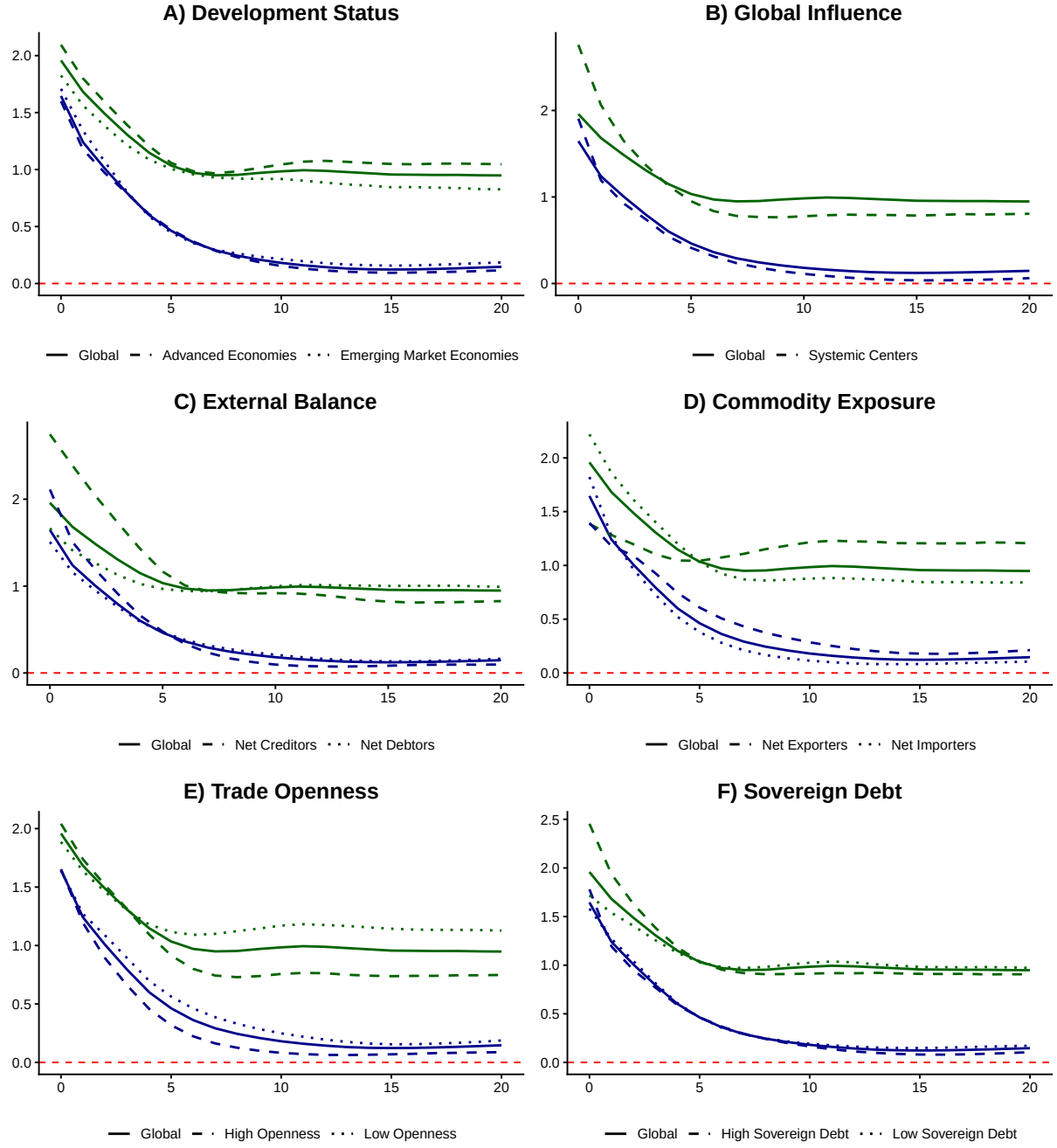


Figure 3: The impact of structural characteristics on the state-dependent sacrifice ratios.

Notes: Solid lines reproduce the global median sacrifice ratio from Figure 2B for the high- (green) and low- (blue) uncertainty regimes; dashed and dotted lines show the corresponding median sacrifice ratios for each country subgroup, obtained by pooling posterior draws within the subgroup rather than across the full sample. Country groupings are defined in the Appendix.

Does the level of uncertainty matter? Figure 4 checks whether the result is an artifact of the specific threshold used to separate high- from low-uncertainty periods. Apparent

from the sacrifice ratios reported here, it is not. The median, 70th-percentile, and cyclical (above-trend) classifications produce sacrifice ratio paths where high uncertainty pushes the sacrifice ratio up. The level at which high uncertainty is defined only appears to matter at the margin. Pushing the threshold level up from the median to the 70th percentile, for instance, somewhat raises the sacrifice ratio at almost all horizons, indicating that possibly the higher the level of uncertainty is, the worse the inflation-output tradeoff becomes for the central bank.

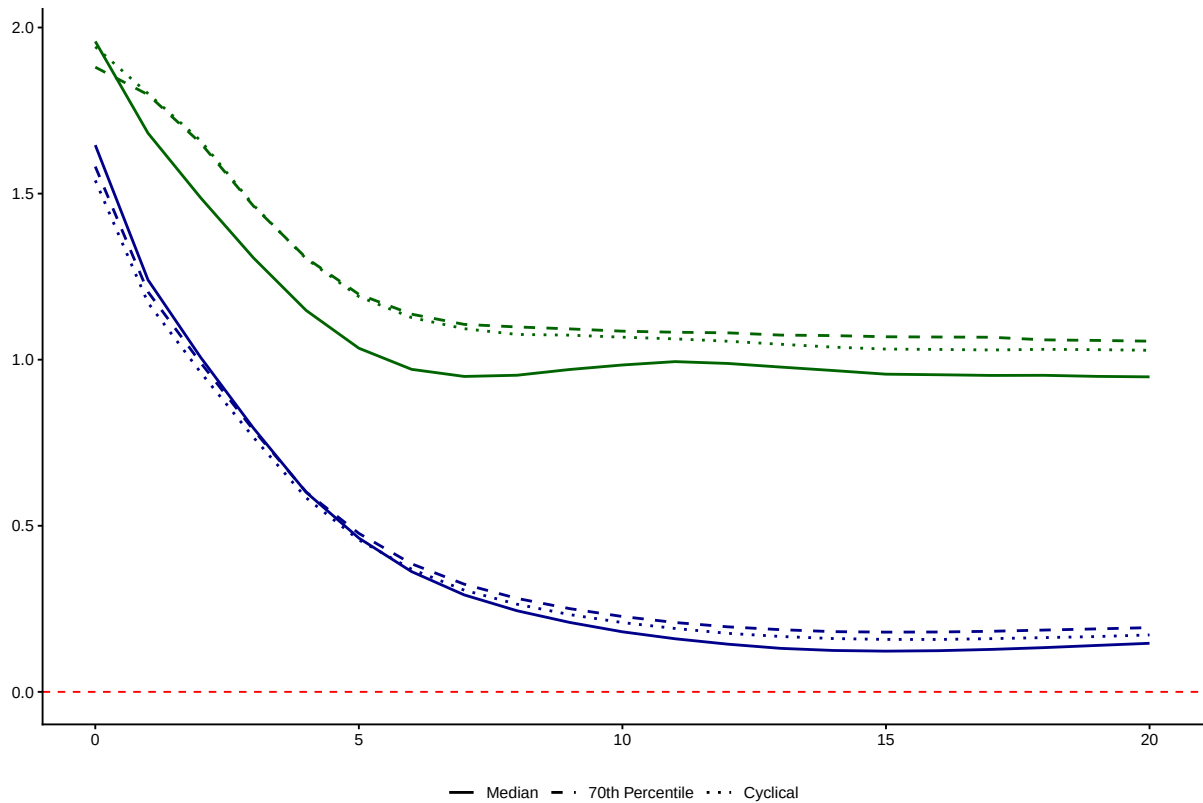


Figure 4: State-dependent sacrifice ratios under alternative threshold definitions.

Notes: Solid lines show the baseline median-threshold classification from Figure 2B; dashed and dotted lines show the corresponding sacrifice ratios when the high-uncertainty regime is instead defined as periods above the 70th percentile, or as periods above trend ("cyclical"), of the relevant uncertainty index.

How sensitive are the results to the uncertainty measurement? The result is not uniform across uncertainty measures. Figure 5 shows that the headline pattern is clearest for the two narrative, text-based measures of policy-type uncertainty, the WUI and the

EPU. It is considerably weaker under the JLN1M index, and reverses under the GPR index, where the low-uncertainty regime shows the higher sacrifice ratio instead.

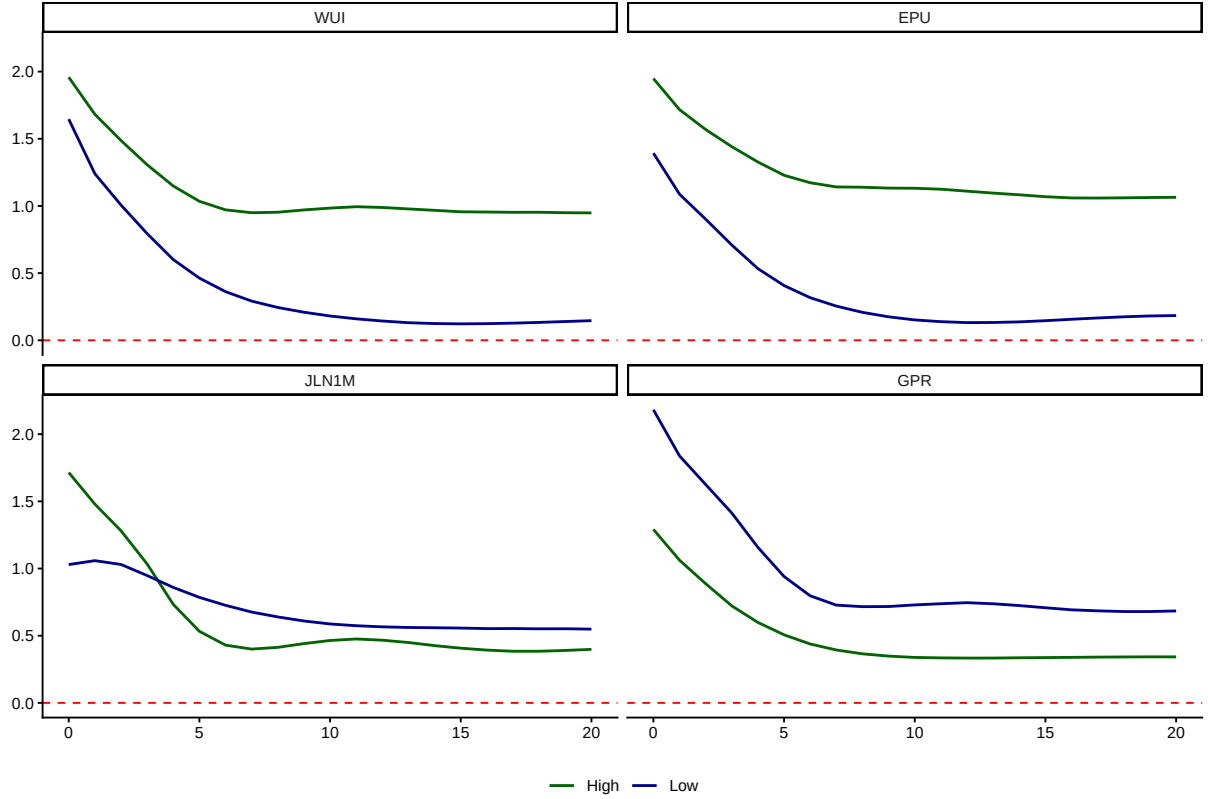


Figure 5: State-dependent sacrifice ratios under alternative uncertainty measures.

Notes: Each panel replaces the baseline WUI-based regime classification with the indicated alternative uncertainty measure (EPU, JLN1M, or GPR; see Data section) and re-estimates the model accordingly. Green and blue lines denote the high- and low-uncertainty regimes as defined by that measure.

Figure 11 in the Appendix points to where this divergence comes from. Under WUI and EPU in the low-uncertainty regime, the inflation responses' deep and persistent drop is what drives the low sacrifice ratio seen in Figure 2. Under JLN1M and GPR, the low-uncertainty inflation response closes markedly faster, practically the relationship here swaps. Under these two indices, there appears to be more of a difference between output responses too. For the GPR index, in low uncertainty times, the real contraction is largest out of all specifications and considerably larger relative to the high uncertainty counterpart with the same index. Regarding the JLN1M index, while the initial drop is larger in

high uncertainty, output adjusts quicker, whereas the transmission to real output in low uncertainty times indicates a slightly more prolonged response. With respect to output, these results resonate more with the findings of earlier papers.

This is consistent with the conceptual distinctions already drawn in the Data section. JLN1M is derived from realized macro-financial forecast-error volatility rather than perceived policy uncertainty, and GPR captures geopolitical tension specifically rather than economic-policy uncertainty. Read together, the result may be better understood as a finding about policy-type uncertainty in particular. While the results may not hold under all sources of uncertainty, hesitancy in price-setting driven by unresolved questions about the future policy and regulatory environment does appear to hinder the effectiveness of monetary policy in achieving disinflation.

Robustness to the identifying restrictions. In order to test the robustness of our results, we drop the identifying restrictions imposed on horizons $h = 1, 2$ and rely only on the $h = 0$ restrictions to identify the monetary policy shock. Figure 6 shows that the implications of the results hold even when relaxing the identifying restrictions. While on shorter horizons the sacrifice ratios coincide, they stabilize such that the high-uncertainty ones are consistently above the low-uncertainty ratios. Figure 12 in the Appendix shows where the sensitivity that does exist is concentrated. Relaxing the restriction primarily changes the GDP responses, while the inflation impulse response patterns stay largely unchanged regardless of identifying restrictions. Since inflation has been the primary driver of our results, this reassures us that the core conclusions are unchanged.

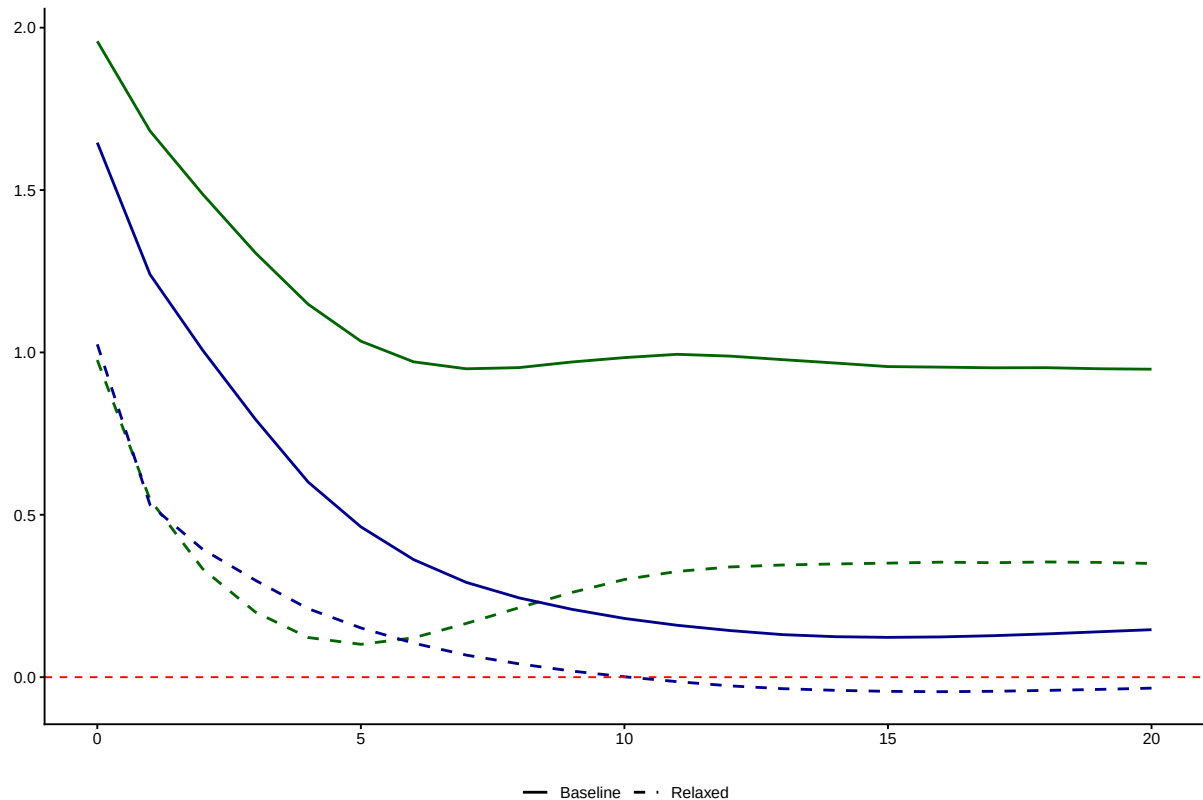


Figure 6: Robustness check: relaxing the persistence restriction on the monetary policy shock.

Notes: "Baseline" imposes the sign restrictions in Tables 1 and 2, including persistence of the monetary policy shock through $h = 1, 2$; "Relaxed" drops the horizon-1–2 persistence restriction on the monetary policy shock while retaining the contemporaneous ($h = 0$) restrictions.

Conclusion

This paper set out to ask whether, and how, economic uncertainty alters the transmission of monetary policy. Using Bayesian Threshold VARs estimated across a broad panel of 30 advanced and emerging market economies, with monetary shocks identified via sign restrictions and the full system allowed to vary by uncertainty regime, we find a clear answer. Transmission is state dependent, but not through the channel most of the existing literature emphasizes. The response of GDP growth to a monetary policy shock is broadly similar whether uncertainty is high or low. What differs sharply is the response of inflation, which is shallower and closes faster in the high-uncertainty regime than in

the low-uncertainty regime. The resulting sacrifice ratio is several times larger under high uncertainty, making disinflation a considerably more expensive undertaking for central banks precisely when uncertainty is elevated.

Two features of the result stand out on closer inspection. First, it is inflation, not output, that carries the state dependence, which reframes the policy problem. Uncertainty does not so much prevent a central bank from contracting the economy as it prevents that contraction from buying disinflation. We interpret this, tentatively, as most consistent with an aggregate-supply channel (a flattening Phillips Curve) though a reduced-form, sign-restricted framework cannot distinguish this from an observationally similar story in which less-anchored or harder-to-update inflation expectations play the same role, and we present both as candidate explanations rather than adjudicate between them. The modest state dependence we also find in the policy rate response itself suggests that the systematic conduct of policy, not only the private-sector side of transmission, may adjust under uncertainty as well.

Second, the result is not uniform across uncertainty measures: it is strongest for narrative, text-based indicators of economic policy uncertainty specifically, and weaker or reversed for a model-based macro-uncertainty index and a geopolitical risk index. This suggests the finding is less about uncertainty in a universal sense than about economic policy-related uncertainty interacting with how prices and expectations respond to demand conditions.

Policy implications. Three implications follow. First, and most directly, central banks may need to expect a materially worse inflation-output tradeoff during episodes of elevated policy uncertainty. A given monetary tightening buys less disinflation, so achieving a given inflation objective may require either a larger or a more sustained policy response than the same objective would require in calmer times. Following the logic of the sacrifice ratios we computed, the real economic cost associated with achieving disinflation rises significantly.

Second, because the effect we document is considerably stronger for measures of economic policy uncertainty than for uncertainty in general, our results point to a complementarity between monetary policy effectiveness and the predictability of the broader policy environment (not only monetary policy, but fiscal, regulatory, and trade policy as well). To the extent that episodes of geoeconomic fragmentation of the kind discussed in the introduction generate exactly this type of policy uncertainty, they may impose a cost on inflation

control that falls on central banks even when the underlying disturbance originates well outside their mandate. This is, in effect, an argument for policy coordination and clear communication across government as a complement to monetary policy, not only a matter for central banks acting alone.

Third, if the anchoring-based interpretation of our results has merit, central bank communication and credibility take on added importance precisely when uncertainty is high, since well-anchored expectations are what would prevent a given shock from propagating further into future inflation. This reinforces a familiar prescription that investing in credibility matters. With our results in mind, there is a measurable cost of failing to do so.

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Appendix

Supplementary graphs for Figure 2

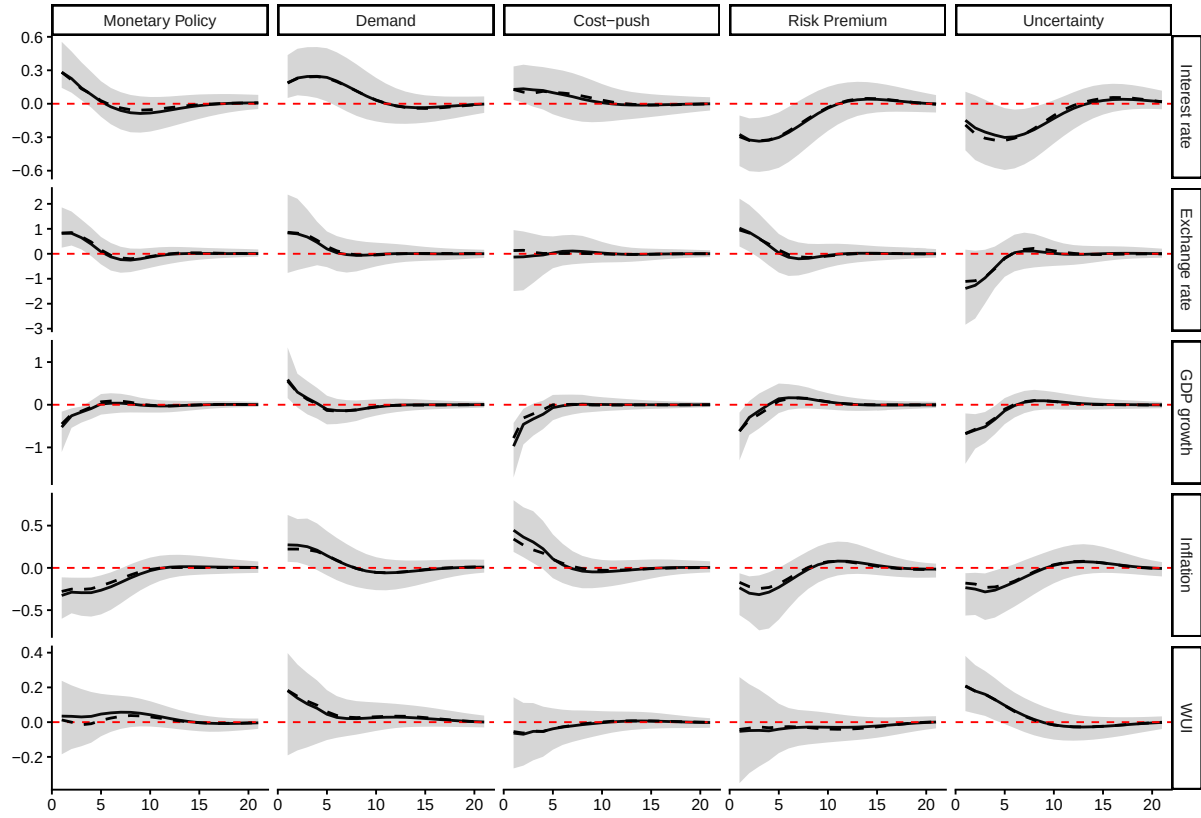


Figure 7: Linear VAR impulse responses to the five identified shocks.

Notes: Solid lines show the median impulse response for the full country panel; dashed lines indicate the median impulse response computed from the subset of economies where estimated shadow short rates are used.

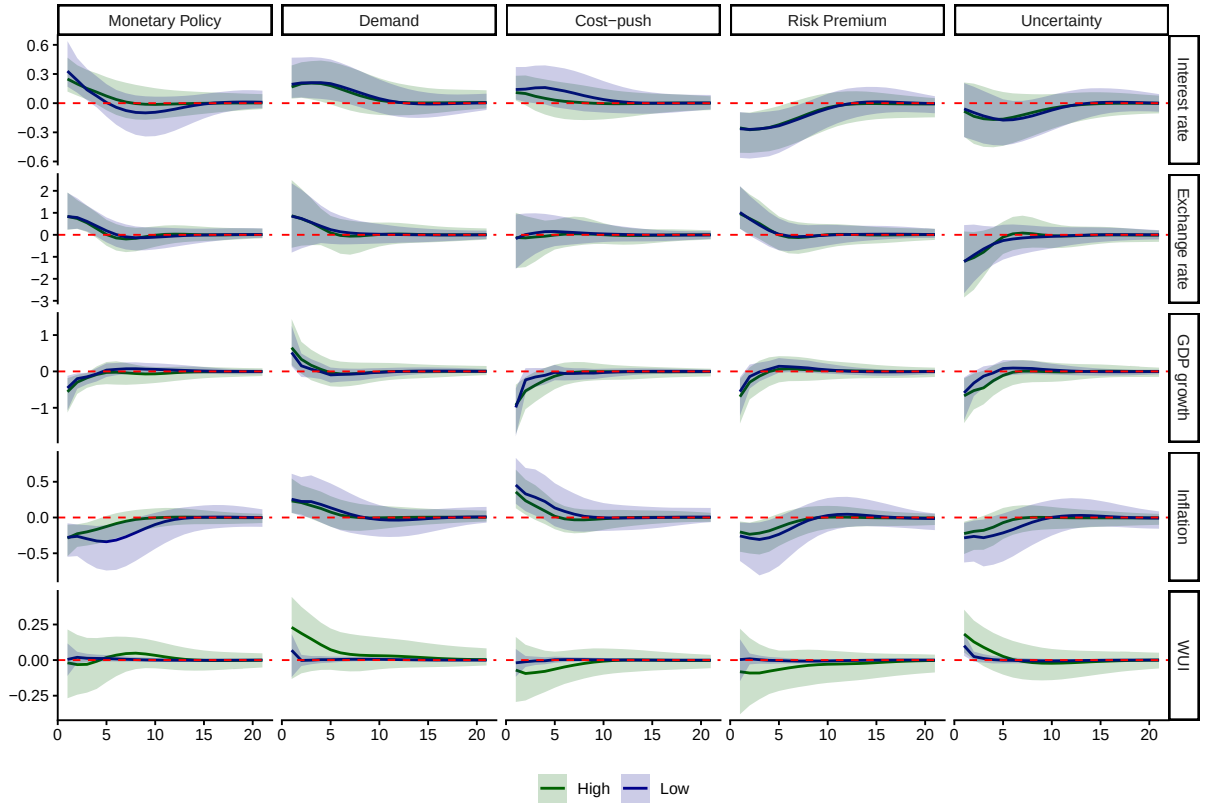


Figure 8: State-dependent impulse responses to the five identified shocks - full sample.

Notes: Green (High) and blue (Low) lines denote posterior median impulse responses in the high- and low-uncertainty regimes; shaded bands show 68% credibility sets. Estimated on the full country panel.

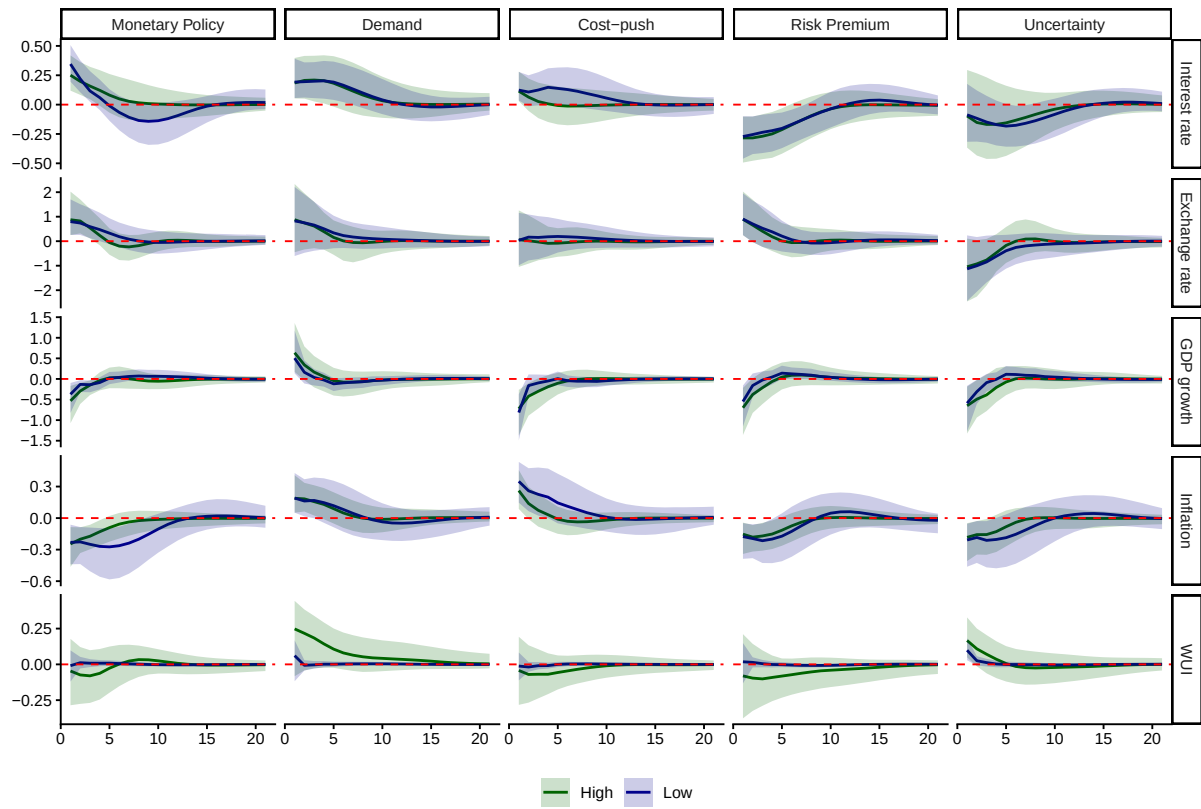


Figure 9: State-dependent impulse responses to the five identified shocks - shadow-rate subsample.

Notes: As Figure 8, restricted to the subset of economies for which estimated shadow short rates are used in place of the observed policy rate (see Data section).

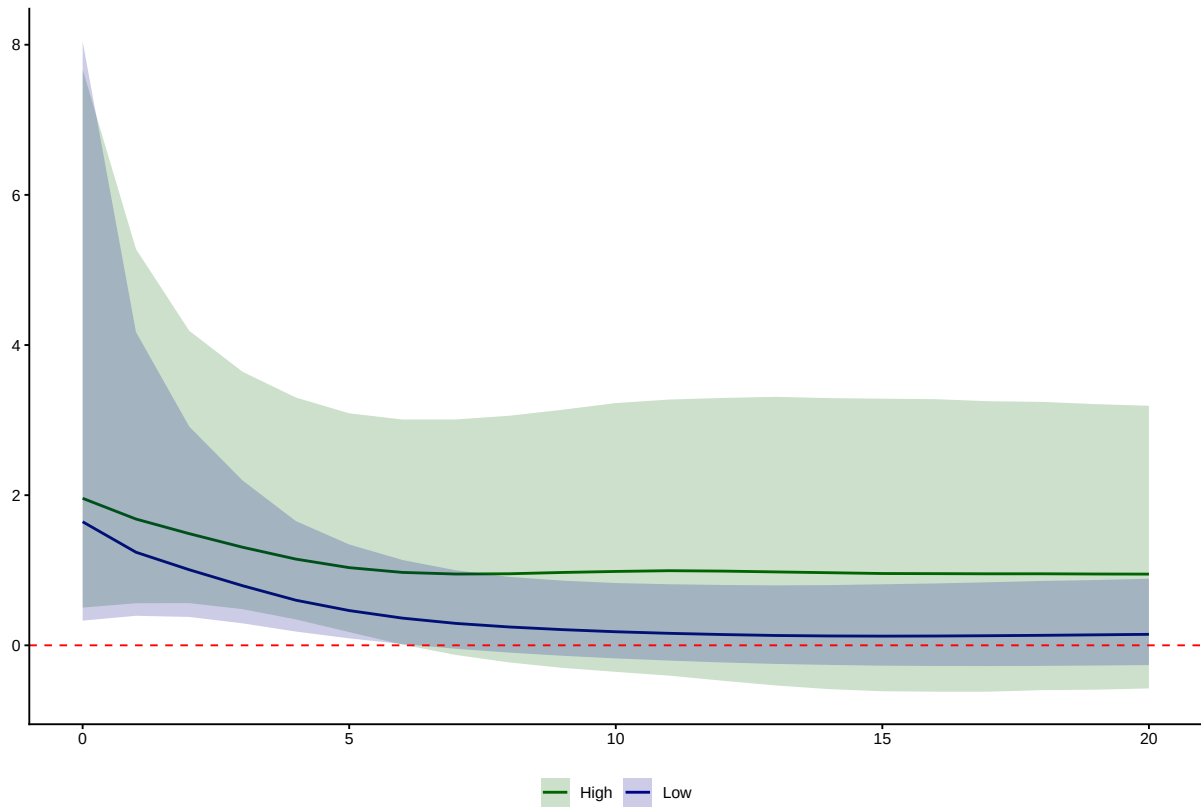


Figure 10: Sacrifice ratios with 68% credibility sets.

Notes: As Figure 2B, but showing the posterior 68% credibility set (shaded) around the median sacrifice ratio (solid line) for each regime.

Classification of country groups in Figure 3

This subsection of the Appendix details the macroeconomic classifications used to partition the sample of 30 economies into the structural groupings presented in Figure 3. The assignments rely on standard institutional databases and established empirical thresholds to ensure consistent cross-country comparisons.

Development Status: Economies are classified as Advanced Economies (AEs) or Emerging Market Economies (EMEs) in accordance with the IMF World Economic Outlook taxonomy. The AE subsample comprises AUS, CAN, CHE, DNK, EA, GBR, ISL, ISR, JPN, KOR, NOR, NZL, SGP, SWE, and USA. The remaining economies are designated as EMEs (BRA, CHL, COL, CZE, HUN, IDN, IND, MEX, MKD, PHL, POL, ROU, SRB, THA, ZAF).

Global Influence: We select a handful of “Systemic Centers” based on gross economic footprint and global financial influence. Systemic Centers include major reserve currency issuers and exceptionally large domestic markets (BRA, EA, GBR, IND, JPN, USA).

Commodity Exposure: Economies are grouped by their structural net trade position in primary commodities. Net Commodity Exporters (AUS, BRA, CAN, CHL, COL, IDN, NOR, NZL, ZAF) are defined as those generating positive net exports of raw materials and energy products. All remaining economies are classified as Net Importers.

External Balance: This classification hinges on the Net International Investment Position (NIIP) to separate Creditor and Debtor economies. Net Creditors possess a structurally positive NIIP (CHE, DNK, EA, ISR, JPN, KOR, NOR, SGP, SWE). The designated Net Debtors operate with a negative NIIP (AUS, BRA, CAN, CHL, COL, CZE, GBR, HUN, IDN, IND, ISL, MEX, MKD, NZL, PHL, POL, ROU, SRB, USA, ZAF).

Trade Openness: Economies are partitioned based on their gross trade-to-GDP ratios utilizing standard trade metrics $\frac{Exports+Imports}{GDP}$. Economies exceeding a high structural openness threshold are classified as High Openness (CHE, CZE, DNK, HUN, ISL, KOR, MKD, POL, ROU, SGP, SRB, SWE, THA), with the remaining economies categorized as having Low to Moderate Openness.

Sovereign Debt: Sovereign debt groupings are determined by general government gross debt-to-GDP ratios. Economies exhibiting elevated structural debt levels are assigned to the High Debt category (BRA, CAN, EA, GBR, HUN, IND, ISR, JPN, USA, ZAF). All remaining economies are classified as having Low to Moderate Sovereign Debt.

Supplementary graphs for the uncertainty measure variation

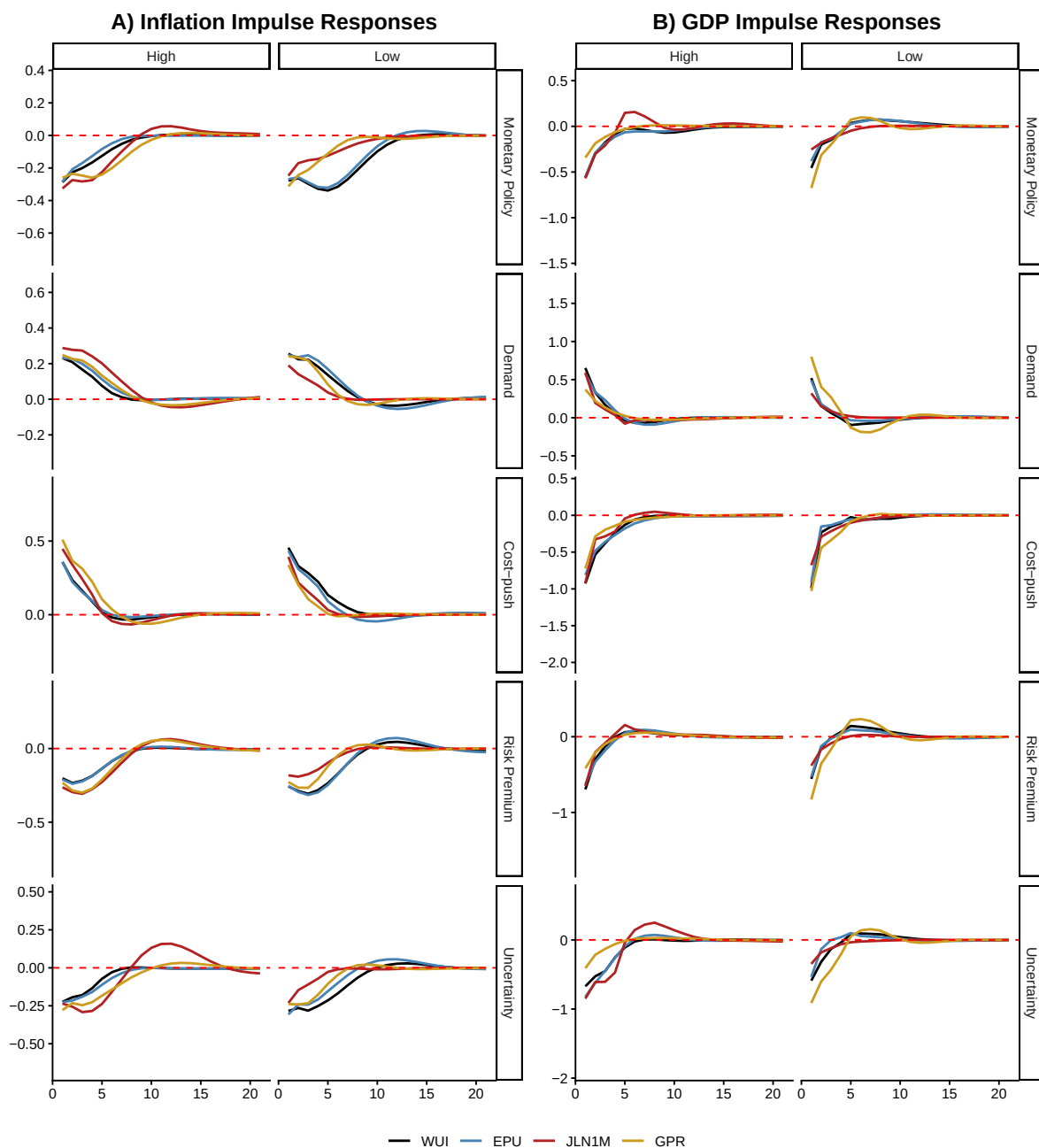


Figure 11: State-dependent GDP and inflation impulse responses under alternative uncertainty measures.

Notes: Each colored line shows the posterior median impulse response under the indicated uncertainty measure (WUI, EPU, JLN1M, or GPR), separately for the high- and low-uncertainty regimes as classified by that measure.

Supplementary graphs for testing the identifying restrictions

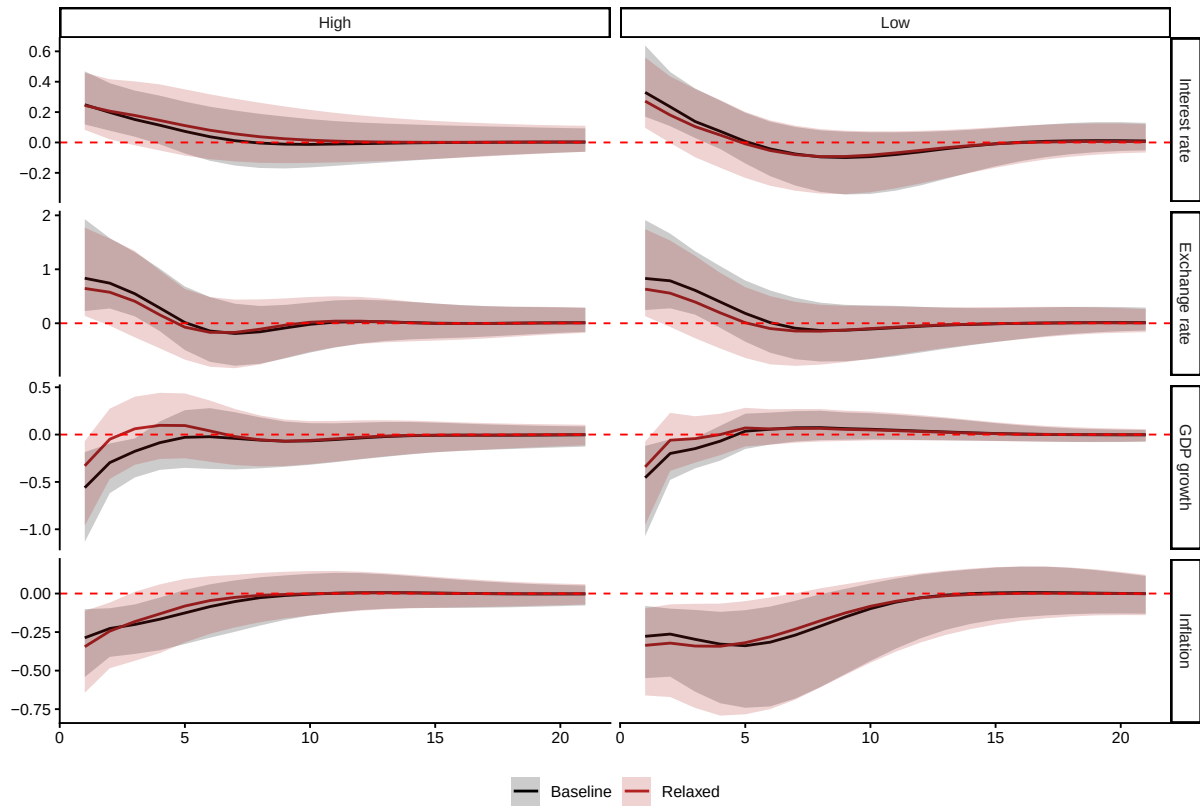


Figure 12: State-dependent impulse responses under the relaxed identifying restriction.

Notes: "Baseline" and "Relaxed" correspond to the two specifications in Figure 6; shaded bands show 68% credibility sets. High- and low-uncertainty regime results are shown in separate columns rather than overlaid, unlike Figures 8 and 9.